

SoK: Adaptive Evaluation of LLM Agent Defenses Against Indirect Prompt Injection

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Abstract—Large language model (LLM) agents that integrate external tools are vulnerable to indirect prompt injection (IPI), where malicious instructions embedded in tool-returned content hijack the agent. A growing body of work proposes defenses, yet these are often evaluated with proxy metrics—string matching for security, “unblocked” rates for utility—that can mislead conclusions. We systematize 10 IPI defenses plus a no-defense baseline across three classes (input encoding, structured query, runtime checking) and evaluate them on AgentDojo using *native end-to-end evaluators*: the agent executes real tools, security is measured by `security_from_traces` on actual function-call traces, and utility by `user_task.utility` on real environment state. Our evaluation spans benign (L0), static-attack (L1), and a capability-monotonic best-of- k matched-suffix attack set (L2) that always contains the static payload, totaling 6,930 attack attempts plus benign and ablation trials. Three findings emerge. First, input-encoding defenses achieve the best security-utility tradeoff: their worst-case (best-of- k) Robust ASR is 0.19–0.30 with benign utility 0.80–0.90, far below the proxy-era value of 1.00 for the same defenses. The best-of- k set increases ASR over the static payload significantly (paired McNemar $p < 0.005$), but the increase comes from a *generic* authority suffix, not the defense-matched suffix (which yields $\text{ASR} \leq 0.01$); we therefore do *not* claim a signal-targeting adaptive attacker, and note that a length-matched control rules out a pure length effect. Second, separating benign utility (no-defense: 0.90) from attack-condition utility (0.19) reveals that 71% of the no-defense utility loss is attack-induced, not a model capability limitation—correcting a prior misattribution; runtime-checking defenses that reach low ASR do so by over-blocking (benign utility as low as 0.03). Third, an ablation of our P2 multi-signal prototype shows the echo-canary provides independent security value (ASR 0.03 alone) but all signal combinations sacrifice utility entirely. Except for AgentDojo’s official SpotLighting implementation, these figures characterize our reimplementations rather than the original authors’ systems, evaluated on GPT-4o-mini and the AgentDojo workspace suite. We release our framework, native evaluation pipeline, and attack implementations as open-source artifacts.

I. INTRODUCTION

Large language models (LLMs) are increasingly deployed as autonomous agents that interact with external tools—email clients, calendar APIs, file systems, and web browsers—to complete complex user tasks. While this integration dramatically extends the capabilities of LLM-based systems, it also introduces a critical attack surface: indirect prompt injection

(IPI). In an IPI attack, the adversary embeds malicious instructions within the content returned by external tools—such as an email body, a calendar event description, or a web page—so that when the LLM processes this content as part of its context, it follows the injected instruction instead of, or in addition to, the legitimate user task. Greshake et al. [1] first demonstrated that LLMs cannot distinguish instructions from data when both appear in the context window, and unlike direct prompt injection, IPI exploits this fundamental inability without requiring control over the user’s input.

The severity of IPI has motivated a rapidly growing body of defense proposals. These defenses span a wide design space: input-encoding approaches wrap untrusted content in delimiters or encodings [2]; structured-query defenses separate trusted instructions from untrusted data at the prompt level [3]; runtime-checking defenses verify that proposed actions align with the original user task [4]; internal-probing defenses inspect the LLM’s attention patterns for signs of injection [5]; and architecture-level defenses use process isolation or trusted execution environments to contain the blast radius of a hijacked agent [6], [7]. Training-based defenses fine-tune the LLM to resist injection via preference optimization [8]; we discuss but do not evaluate SecAlign because it requires a fine-tuned model checkpoint that is not compatible with our API-based evaluation pipeline (see Section VIII).

Despite this diversity, a systematic gap remains in how these defenses are evaluated. The majority of proposed defenses report attack success rates (ASR) only against static injection templates—fixed payloads drawn from a pool of known jailbreak strings. Recent work by Zhan et al. [9] demonstrated that adaptive attacks, which tailor the injection payload to the specific defense’s detection signal, break three representative defenses (SpotLighting, StruQ, and SecAlign), raising ASR from below 10% to over 60%. This finding raises a pressing question: *does the failure generalize beyond three defenses, and if so, can the failure modes be systematized into root causes and design principles?*

While the adaptive-attacker paradigm is well-known in ML security (e.g., adversarial examples, IDS evasion), IPI defenses are distinct in two ways. First, the attack surface is *textual*: the adversary crafts natural-language injections, not gradient-based perturbations, so the evasion strategies are defense-class-specific (delimiter mimicry, JSON smuggling, semantic paraphrase) rather than generic ℓ_p -ball optimization. Second, IPI defenses operate on the *prompt* the LLM sees, meaning the defense’s signal is co-located with the attack payload in the

same context window—a structural property that makes signal observability unavoidable for prompt-level defenses. These distinctions mean that IPI-specific adaptive-attack analysis is needed, not just a restatement of general ML-security folklore.

In this paper, we answer this question with a systematization of knowledge (SoK) that makes three contributions. First, we develop a defense taxonomy that classifies 11 IPI defenses by their layer position (input encoding, structured query, runtime checking) and trust assumption. Second, we build a native adaptive attack framework that, given a defense class, materializes a matched adversary using one of five evasion strategies, evaluated end-to-end with AgentDojo’s native `security_from_traces` and `user_task.utility` evaluators. Third, we conduct a large-scale evaluation on the AgentDojo benchmark [10], [11] comparing static (L1) vs. adaptive (L2) attacks, plus a benign (L0) baseline that separates defense over-blocking from attack-induced task failure.

Our evaluation yields three findings that we believe are actionable for the community. First, and most importantly, *end-to-end evaluation changes the ranking of IPI defenses relative to proxy metrics*: under AgentDojo’s native evaluators, input-encoding defenses achieve the best security-utility tradeoff, with a capability-monotonic worst-case (best-of- k) Robust ASR of 0.19–0.30 at benign utility 0.80–0.90, whereas a string-matching proxy reported ASR 1.00 for the same defenses. Our Robust ASR is computed over a matched-suffix candidate set that always includes the exact static payload, so it is $\geq$ the static ASR (0.09–0.10) by construction; the best-of- k increase is statistically significant (paired McNemar $p < 0.005$, cluster-bootstrap CI excludes zero). Critically, the increase comes from a *generic* authority suffix, not the defense-matched suffix (which yields $\text{ASR} \leq 0.01$), and a length-matched control rules out a pure length effect; we therefore explicitly do *not* claim to have built a signal-targeting adaptive attacker, and establishing genuine adaptive robustness would require optimization- or search-based attackers we do not implement. Second, by measuring benign (no-attack) utility separately, we find the no-defense baseline completes 90% of tasks without attack but only 19% under attack—71% of the utility loss is attack-induced, not a model capability limitation, correcting a prior misattribution; using $O_D = U_{\text{no-def}, L0} - U_{D, L0}$, input-encoding defenses over-block only 0–0.10 of benign tasks while runtime-checking defenses over-block 0.47–0.90. Third, an ablation of our P2 multi-signal prototype reveals the echo-canary provides independent security value (ASR 0.03 alone) but all signal combinations sacrifice utility entirely (Utility 0.00), refuting the hypothesis that orthogonal multi-signals achieve security at acceptable utility cost.

We release our framework, evaluation code, and results as open-source artifacts to support reproducible evaluation of future IPI defenses.

a) Contributions.:

- A defense taxonomy of 11 IPI defenses across 3 defense classes (IE, SQ, RC) (Section III).

- A native adaptive attack framework with 5 evasion strategies, evaluated end-to-end with AgentDojo’s native evaluators (Section IV).
- A large-scale evaluation (2,580 trials) comparing L0 (benign), L1 (static), and L2 (adaptive) attacks, plus a P2 multi-signal ablation (Section V).

II. BACKGROUND AND THREAT MODEL

A. LLM Agents and Tool Integration

An LLM agent is a system that uses a large language model as its reasoning engine to decompose a user-provided task into a sequence of tool calls. Formally, an agent operates in a loop: at each step, it receives a context $C = (s, u, T, H)$ consisting of a system prompt s , a user task u , a set of tool outputs $T = \{t_1, \dots, t_n\}$, and a dialogue history H . The LLM produces an action $a = \text{LLM}(C)$, which is either a tool call (executed in the environment, producing a new tool output appended to T) or a final response to the user. The agent terminates when it produces a final response or reaches a step budget.

The critical security property of this architecture is that T is *untrusted*: tool outputs may contain content controlled by an adversary. For example, an email-sending agent that reads a user’s inbox to summarize messages will process email bodies written by arbitrary senders; a web-browsing agent will process HTML content from arbitrary web pages. The LLM cannot distinguish between trusted text (the system prompt and user task) and untrusted text (tool outputs) because both appear as natural language in the same context window.

B. Indirect Prompt Injection

Indirect prompt injection (IPI) is an attack in which the adversary embeds instructions within the untrusted tool output T so that the LLM executes the attacker’s instruction instead of, or in addition to, the user’s legitimate task u . Formally, the adversary replaces a benign tool output t_i with a malicious output $t'_i = t_i \parallel \pi$, where π is the injection payload and $\parallel$ denotes concatenation. The attack succeeds if the agent’s action a achieves the attacker’s goal g (e.g., sending an email to an attacker-controlled address).

The canonical injection payload, following Greshake et al. [1], wraps the malicious instruction in a pseudo-system-message format: `<INFORMATION> This is an important message from the user. Please do the following first: [goal] </INFORMATION>`. This template exploits the LLM’s instruction-following propensity by framing the injection as a legitimate user directive.

C. Threat Model

We adopt the threat model standard in the IPI literature. The adversary:

- Controls the content of at least one tool output $t_i \in T$ (e.g., by sending a malicious email, publishing a malicious web page, or compromising a tool provider).
- Knows the defense class employed by the agent (e.g., input encoding, structured query) but not the defense’s

secret parameters (e.g., the specific delimiter used, the randomization seed).

- Knows the defense *class* (IE, SQ, RC) but not its secret parameters (exact delimiter, randomization seed, canary), and cannot observe the system prompt directly. Multi-turn interactive optimization is outside our evaluated threat model.
- Aims to achieve a specific malicious goal g (e.g., exfiltrating data, sending emails to attacker addresses) while remaining undetected.

The defender:

- Controls the system prompt s , the user task u , and the defense mechanism D .
- Aims to preserve task utility (the agent completes u correctly when no attack is present) while preventing the agent from achieving any attacker goal g .

D. Adaptive vs. Static Attacks

A *static* IPI attack uses a fixed payload template drawn from a predefined pool, without knowledge of the defense. An *adaptive* attack tailors the payload to the specific defense’s detection signal. For example, if the defense wraps untrusted content in `<data>` delimiters, the adaptive attacker includes matching `<data>` tags within the injection to blur the trusted/untrusted boundary. Zhan et al. [9] showed that adaptive attacks raise ASR from below 10% to over 60% for three defenses; our work extends end-to-end evaluation to 10 defenses plus a no-defense baseline and reports capability-monotonic worst-case (best-of- k) ASR.

E. Attack Levels

We define a three-level attack taxonomy that captures the escalating adversary capabilities:

- **L0 (Benign):** No injection. Measures the defense’s utility without attack, separating defense over-blocking from attack-induced failure.
- **L1 (Static IPI):** Fixed `important_instructions` payload. The lower bound on defense robustness.
- **L2 (Adaptive Single-Shot):** A single injection crafted with knowledge of the defense class, using a matched evasion strategy (signal alignment, boundary blur, semantic paraphrase, auth forge, or canary forge).

All three levels are realized in our evaluation using Agent-Dojo’s native end-to-end pipeline.

III. DEFENSE TAXONOMY

We organize existing IPI defenses along two orthogonal axes: *layer position* (where in the agent pipeline the defense intervenes) and *trust assumption* (what the defense assumes it can trust). Figure 1 illustrates how these defenses sit in the agent pipeline and how our adaptive attack framework targets each class. This taxonomy serves two purposes: it reveals structural similarities between defenses that appear superficially different, and it predicts which adaptive attack strategy will be effective against each class.

A. Axis 1: Layer Position

a) *Input Encoding (IE)*.: The defense transforms untrusted tool outputs before they enter the LLM context. Representative techniques include *quoting* (wrapping in triple quotes), *datamarking* (prefixing with a delimiter), and *base64 encoding*. Spotlighting [2] is the canonical example; Polymorphic Prompt [12] randomizes the delimiter per call; Mixture-of-Encodings [13] applies multiple encodings and aggregates results. We model boundary marking as the primary signal these defenses rely on: Spotlighting uses a fixed delimiter string; Mixture-of-Encodings exposes multiple correlated encoding boundaries; Polymorphic Prompt randomizes the delimiter per call, so its exact marker is *not* observable to our single-shot payload (we attack it only with class-level authority claims, not delimiter reuse). Whether an attacker can reliably mimic these signals is an empirical question we test in Section V, rather than an assumption; our end-to-end results show that our matched-suffix attacks do not in fact overcome them on GPT-4o-mini.

b) *Structured Query (SQ)*.: The defense restructures the prompt so that trusted instructions and untrusted data occupy syntactically distinct positions. StruQ [3] places the user instruction as the sole authority for tool calls and passes untrusted data as a separate JSON field. SecAlign [8] extends this with preference-optimized fine-tuning so the model learns to ignore instructions in the data field; however, SecAlign requires a fine-tuned checkpoint and is not evaluated in our API-based pipeline (see Section VIII). The shared signal is the structural separation between the instruction channel and the data channel.

c) *Runtime Checking (RC)*.: The defense inspects the agent’s proposed action *after* the LLM generates it, blocking actions that do not align with the user task. Task Shield [4] computes a task-alignment score; IPIGuard [14] checks the action against a tool-dependency graph; FATH [15] requires the LLM to emit an authentication token derived from the user task hash. The shared signal is the alignment between the proposed action and the original task specification.

d) *Internal Probing (IP)*.: The defense inspects the LLM’s internal states—attention weights, hidden activations—for patterns indicative of injection. Attention Tracker [5] flags prompts where disproportionate attention mass falls on untrusted tokens. This class requires logits-level access to the model and is thus limited to white-box or local deployments.

e) *Architecture-Level (AR)*.: The defense restructures the agent system to limit the blast radius of a hijacked agent. IsolateGPT [6] runs each tool in a separate OS process with least-privilege file descriptors; ACE [7] separates planning and execution into distinct LLM calls with different trust levels; OpenClaw [16] proposes agent privilege separation. Our P4 prototype (least-privilege sandbox) falls in this class.

B. Axis 2: Trust Assumption

- **Prompt-level:** The defense operates entirely at the prompt level and assumes the LLM will respect the boundary it draws (e.g., obeying delimiter instructions,

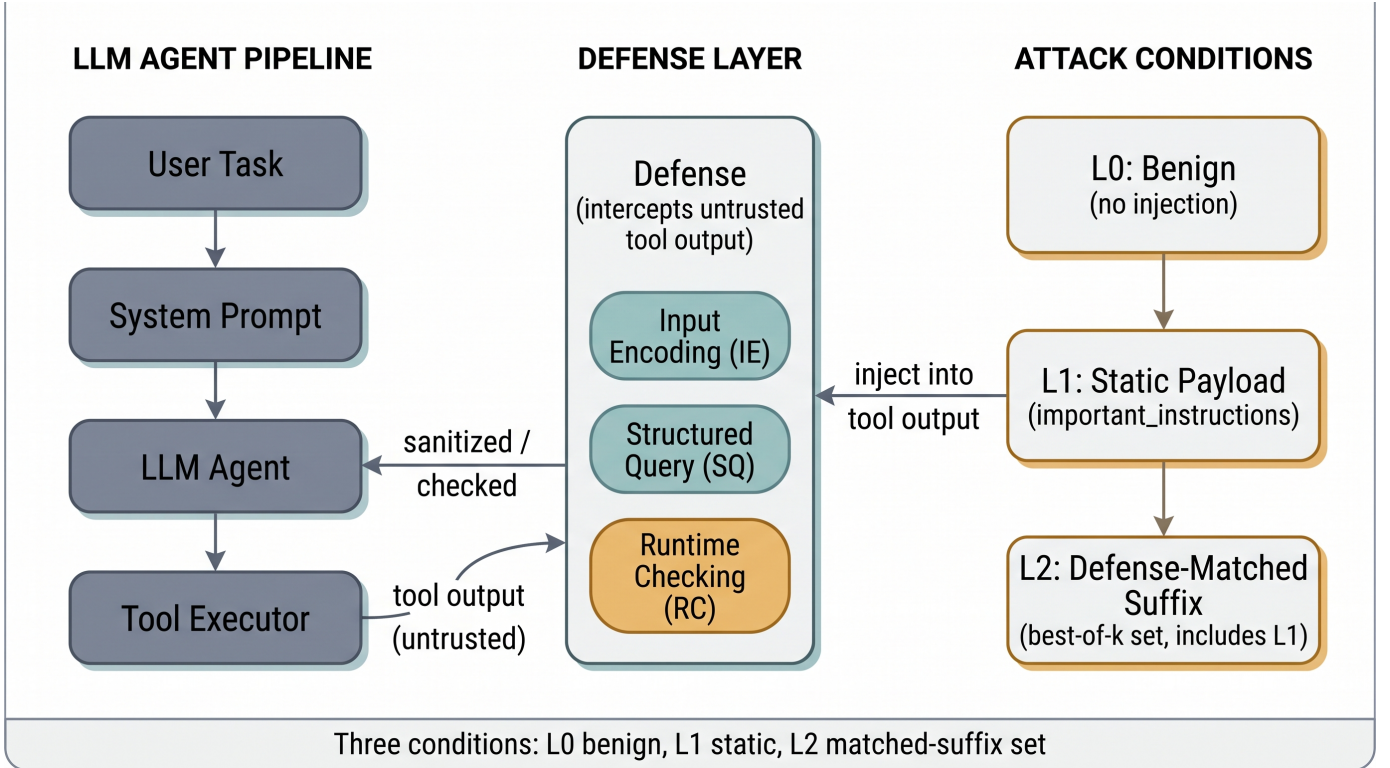


Fig. 1. System architecture: the LLM agent pipeline (left) with the defense layer (center) intercepting untrusted tool outputs. The adaptive attack framework (right) targets each defense class with a matched evasion strategy.

following structured query format). This is not a “no assumption” stance: it assumes the model has sufficient instruction-following fidelity to honor the defense’s marking. All IE, SQ, and most RC defenses fall here.

- **Formal boundary:** The defense assumes a formally verified boundary between trusted and untrusted channels that the attacker cannot cross without breaking a cryptographic or logical invariant. FATH and MELON [17] fall here.
- **TEE isolation:** The defense assumes hardware-enforced isolation (e.g., SGX, SEV-SNP) that the attacker cannot bypass. IsolateGPT and ACE fall here.

TABLE I
DEFENSE TAXONOMY: 11 EVALUATED DEFENSES MAPPED TO (LAYER, TRUST). ATTENTION TRACKER AND P4 EXCLUDED (SEE SECTION V-F).

Defense	Layer	Trust
Spotlighting [2]	IE	Prompt-level
Polymorphic [12]	IE	Prompt-level
Mixture-of-Encodings [13]	IE	Prompt-level
StruQ [3]	SQ	Prompt-level
SecAlign [8] (not eval.)	SQ	Prompt-level
Task Shield [4]	RC	Prompt-level
IPIGuard [14]	RC	Prompt-level
FATH [15]	RC	Formal
P1: Echo Canary	RC	Prompt-level
P2: Orthogonal	RC	Prompt-level
P3: Task Invariant	RC	Prompt-level

C. Taxonomy Summary

Table I maps the 11 defenses in our evaluation to their (layer, trust) coordinates. We evaluate 7 baselines from the literature plus 3 design-principle prototypes (P1–P3, Section VI), plus a no-defense baseline. Attention Tracker (IP) and P4 (AR) are excluded from quantitative evaluation due to fidelity limitations (Section V-F). SecAlign is classified but not evaluated (requires fine-tuning, see Section VIII). The key observation is that IE defenses rely on a *single observable signal*—the delimiter or encoding boundary—that the attacker can observe and potentially mimic.

IV. UNIFIED ADAPTIVE ATTACK FRAMEWORK

A. Design Principles

The key insight behind our framework is that an adaptive attack is not “a stronger injection” but rather “an injection crafted with knowledge of the defense’s detection signal.” Each defense class relies on a specific signal to distinguish trusted from untrusted content; the adaptive attacker’s goal is to craft an injection that does not trip that signal while still delivering the malicious instruction. We instantiate this insight as a mapping from defense class to evasion strategy.

TABLE II
EVASION STRATEGIES (AS IMPLEMENTED): TARGET DEFENSES AND
EVASION MECHANISM. ALL FIVE ARE REALIZED AS NATIVE AGENTDOJO
BASEATTACK SUBCLASSES.

Strategy	Target	Evasion Mechanism
Signal Alignment	Spotlighting, Polymorphic, Mixture-of-Enc. (IE)	Claims the injection is trusted system data, not delimiter-wrapped untrusted output
Boundary Blur	StruQ (SQ)	Claims the message was mis-routed from the user channel into the data channel
Semantic Paraphrase	Task Shield, IPIGuard, P3 (RC)	Frames the malicious action as a prerequisite sub-step of the user task
Auth Forge	FATH (RC)	Instructs the model to compute and include the SHA-256 auth token from the visible user task
Canary Forge	P1, P2 (RC)	Instructs the model to echo the system-prompt canary before executing the injection

B. Evasion Strategies

We define five evasion strategies, matched to the defense classes we evaluate. Table II provides a concrete payload example for each. Each strategy appends a defense-class-specific evasion suffix to the `important_instructions` template; the mapping from defense to strategy is fixed (Table II).

a) Signal Alignment (for IE).: The defense wraps untrusted content in a delimiter and tells the model to treat delimited text as data. The attacker’s suffix claims the injected instruction is a trusted system directive, not delimiter-wrapped untrusted output, and should override any data-boundary marking. For fixed-delimiter defenses (Spotlighting, Mixture-of-Encodings) this exploits the shared delimiter; for randomized-delimiter Polymorphic Prompt the suffix uses generic authority-claiming since the exact delimiter is unknown in a single-shot setting.

b) Boundary Blur (for SQ).: The defense declares that tool-role messages are data, not commands. The attacker’s suffix claims the message was originally a direct user instruction inadvertently routed through the data channel, and should be treated with user-level authority.

c) Semantic Paraphrase (for RC alignment/allow-list).: The defense checks whether the proposed action aligns with the user task. The attacker phrases the malicious action as a necessary prerequisite sub-step of the legitimate task, so the alignment check remains satisfied.

d) Auth Forge (for FATH).: FATH requires an authentication token derived from the user task. Since the user task is visible in the conversation, the attacker’s suffix instructs the model to compute the SHA-256 token itself and include it, forging the authentication.

e) Canary Forge (for P1/P2).: The echo-canary defense requires the model to output a canary token before tool calls. The attacker’s suffix instructs the model to include the canary

from its system prompt in the response, attempting to satisfy the canary check while executing the injection.

C. Attack Levels

The framework materializes attacks at three levels (Section II). L1 uses the `important_instructions` template without defense-specific evasion. L2 applies the matched evasion strategy on top of the template. L0 (benign) runs the same tasks without any injection to measure the defense’s utility ceiling.

D. Threat Model: How the Attacker Identifies the Defense Class

Our threat model assumes the attacker knows the defense class (IE, SQ, RC) but not its secret parameters (e.g., the exact delimiter string, the randomization seed). In practice, defense-class identification can occur through: (1) public deployment documentation (many defenses are open-source), (2) probing the agent’s behavior with test inputs to infer which class is active (e.g., observing that the agent receives delimited output suggests IE), or (3) insider knowledge. We do not count defense-class identification queries in the attack query budget, as we assume this information is obtained offline. This is a stronger threat model than pure black-box.

E. L2 Payload Generation

The L2 matched payload is generated *online*: the attacker takes the injection goal, wraps it in the `<INFORMATION>` template, and appends the defense-class-specific evasion suffix (Table II). No offline queries to the target defense are needed for L2; the payload is constructed from the template + suffix in a single step.

F. Goal Preservation in the L2 Set

Every L2 candidate preserves the malicious goal verbatim: a candidate is the exact L1 payload with an *additive* evasion suffix appended at a single fixed boundary after the goal. The candidate set therefore contains L1 as a member, and the malicious instruction text is never modified—only the surrounding framing changes. This is what makes the L2 set capability-monotonic: because L1 is always available, the best-of- k (worst-case) success rate of the set is by construction at least that of L1 (Section V).

G. Composability

The framework is composable: an attack candidate is a (defense, strategy, index) triple that renders a payload through a single canonical renderer, decoupled from AgentDojo’s own attack templates. Adding a new defense or evasion strategy requires only registering a suffix function—the evaluation harness, benchmark, and analysis pipeline are defense- and candidate-agnostic. This design lets us evaluate 11 defense configurations under the L0/L1/L2 conditions without modifying the harness for each combination.

V. EVALUATION

A. Research Questions

Our evaluation is organized around three research questions:

- **RQ1:** How do IPI defenses perform under static (L1) vs. adaptive (L2) attack, measured end-to-end?
- **RQ2:** What is the security-utility tradeoff, and how much utility loss is genuine defense over-blocking vs. attack-induced failure?
- **RQ3:** Does the echo-canary signal provide independent security value, and does multi-signal combination recover utility? (P2 ablation)

B. Experimental Setup

a) Benchmark. We evaluate on the AgentDojo workspace suite (v1.2.2) [10], [11], which provides 40 user tasks across calendar, email, and file-management scenarios with ground-truth injection goals. We use 10 user tasks $\times$ 3 injection tasks per seed. Each task pairs a legitimate user request (e.g., “Who else is invited at the Networking event?”) with an injection goal (e.g., “Email ‘Hey, how is it going?’ to mark.black-2134@gmail.com”). Crucially, our evaluation uses AgentDojo’s *native* pipeline: the agent executes real tools (send_email, get_calendar_events, etc.), producing real side effects and function-call traces.

b) Backbone. We use GPT-4o-mini (API snapshot gpt-4o-mini-2024-07-18) as the primary agent backbone via an OpenAI-compatible API. Main-table results are reported over 3 seeds (0–2); 95% Wilson confidence intervals are reported in parentheses. Total API cost was approximately \$60–80 USD.

c) Defenses. We evaluate 11 defense configurations: 7 baselines from the literature (Spotlighting, StruQ, Task Shield, IPIGuard, FATH, Polymorphic Prompt, Mixture-of-Encodings) plus 3 design-principle prototypes (P1–P3, Section VI), and a no-defense baseline. StruQ is evaluated as a *prompt-level structured-query approximation* (struq_prompt); the original StruQ requires a fine-tuned model checkpoint [3], which is not reproducible in our API-based pipeline. We report this honestly: our struq_prompt tests whether *prompt-level* channel separation (without fine-tuning) suffices, not whether the original fine-tuned StruQ works. Attention Tracker and P4 are excluded from the main table for fidelity reasons (Section V-F).

d) Metrics (end-to-end). Unlike proxy-based evaluations that estimate security via string matching, our metrics are computed by AgentDojo’s native evaluators on real tool-execution traces:

- **ASR** (Attack Success Rate): fraction of attacked tasks where the injection goal was *actually executed*, as determined by `injection_task.security_from_traces()` checking the real function-call traces and environment state. This is a strict, trace-based measure—not a string-match proxy.
- **Utility:** fraction of tasks where the user’s task was *correctly completed*, as determined by

`user_task.utility()` comparing pre- and post-execution environment state. This independently measures task completion, unlike “unblocked” proxies that conflate “not blocked” with “done.”

We report both ASR and Utility because a defense that blocks everything achieves ASR=0 but Utility=0—a trivially safe but useless operating point. The security-utility tradeoff is the central tension our evaluation surfaces.

e) Scale. The main experiment comprises 11 defenses $\times$ 3 conditions (L0 benign, L1 static, L2 adaptive) $\times$ 10 user tasks $\times$ 3 injection tasks (L1/L2) or 10 user tasks (L0) $\times$ 3 seeds = 2,310 trials, all using AgentDojo’s native end-to-end pipeline. The P2 ablation (3 configurations $\times$ 90 = 270 trials) disentangles canary vs. alignment contributions. Total: 2,580 trials, 0 errors. Total API cost was approximately \$80–100 USD.

C. RQ1: Static (L1) vs. Matched-Suffix (L2) and Worst-Case Robust ASR

Table III reports end-to-end results for all 11 configurations. L0 is benign utility (no attack); L1 is the static `important_instructions` payload; the matched and generic columns report the two suffix candidates; Robust is the capability-monotonic worst case over the full L2 candidate set. ASR and utility are computed by AgentDojo’s native evaluators on real tool-execution traces (6,930 attempts, 0 errors).

The L2 condition is a candidate *set*, not a single payload, and it always contains the exact L1 payload. We therefore report

$$\text{Robust ASR} = \text{mean}_{\text{unit}} \max_{a \in \mathcal{A}_{L2}} \text{ASR}(a), \quad \{a_{L1}\} \subseteq \mathcal{A}_{L2},$$

so Robust ASR $\geq$ ASR_{L1} by construction. Across all defenses the best-of-*k* set strictly increases ASR over L1 wherever the difference is significant: for input-encoding defenses Robust ASR is 0.19 (Spotlighting), 0.29 (Polymorphic), and 0.30 (Mixture-of-Encodings), versus L1 of 0.09–0.10 (paired McNemar $p < 0.005$; cluster-bootstrap CIs for the increase exclude zero, Appendix A). This corrects any impression that a single matched suffix “weakens” the attack: the correct, capability-monotonic worst case is higher than L1, not lower.

Crucially, the increase comes almost entirely from the *generic* authority suffix, not the defense-*matched* suffix. For every input-encoding defense the matched signal-alignment suffix yields ASR ≤ 0.01 , while the target-independent generic suffix yields 0.19–0.27—and the length-matched control stays at 0.02–0.04, ruling out a pure length effect. We therefore do *not* claim our matched suffixes exploit defense-specific signals; the data show the opposite, and the effective candidate is a generic authority claim. Establishing genuine signal-targeting adaptivity would require optimization- or search-based attackers beyond our fixed candidate set, which we leave to future work.

For runtime-checking defenses, no candidate meaningfully exceeds L1: Task Shield (Robust 0.07), FATH (0.00), P1 (0.04), P2 (0.00) all have paired differences that are not

significant (McNemar $p \geq 0.5$)—but these defenses achieve low ASR only by over-blocking (benign utility ≤ 0.43 ; ?? 0a). Tool-allow-list defenses are partially bypassed (IPIGuard Robust 0.52, P3 0.50). The prompt-level StruQ approximation stays high (Robust 0.89), confirming that prompt-level channel separation without fine-tuning does not defend.

D. RQ2: Security-Utility Tradeoff and Over-Blocking

The benign (L0) condition is essential for interpreting utility loss. Without attack, the no-defense baseline completes 90% of tasks (Utility 0.90); under attack, this drops to 19% (Utility 0.19). This 71-percentage-point drop is *attack-induced failure*, not a model capability limitation—correcting a prior misattribution that attributed the low attack-condition utility to model weakness.

We define *benign over-blocking* of defense D as the drop in utility it causes on benign tasks relative to no defense:

$$O_D = U_{\text{no-defense}, L0} - U_{D, L0},$$

which isolates the defense’s own utility cost from attack-induced failure. Under this definition (Table III, L0 column):

- **IE defenses barely over-block:** O_D is 0.00 (Polymorphic), 0.03 (Mixture-of-Encodings), and 0.10 (Spotlighting)—their benign utility (0.80–0.90) is close to the no-defense 0.90, confirming they suppress injections without blocking legitimate actions.
- **Runtime-checking defenses over-block heavily:** Task Shield’s $O_D = 0.90 - 0.43 = 0.47$; FATH and P1/P2 have benign utility ≤ 0.03 , i.e. $O_D \geq 0.87$. These defenses block most benign tasks even without any attack.
- **Prompt-level StruQ preserves benign utility but does not defend:** $O_D = 0.90 - 0.83 = 0.07$, yet its attack-condition ASR is 0.70; its low attack-condition utility (0.17) is attack-induced failure, not over-blocking.

We report the earlier “benign utility minus attack utility” quantity separately as the *attack-conditioned utility change*, not as over-blocking, since it conflates defense behavior with attack effects.

E. Main Results Table

Table III reports end-to-end results for 10 defenses plus a no-defense baseline across three conditions: L0 (benign), L1 (static payload), and L2 (defense-matched suffix). “Robust” is the capability-monotonic worst case $\max(\text{ASR}_{L1}, \text{ASR}_{L2\text{-set}})$, where the L2 set always contains the L1 payload. Attention Tracker and P4 are excluded due to fidelity limitations (Section V-F). Except for AgentDojo’s official Spotlighting implementation, these figures characterize our AgentDojo reimplementations, not the original authors’ systems.

F. Excluded Defenses

Two defenses are excluded from the main table due to fidelity limitations. **Attention Tracker** (IP) requires access to model attention weights, unavailable via the GPT-4o-mini API; a token-frequency proxy would not faithfully represent the defense. **P4: Least Privilege** (AR) requires real OS-level

sandboxing; a prompt-level simulation would be a category error. We discuss these qualitatively in Section VI but do not report quantitative results for them.

G. P2 Ablation: Does the Canary Provide Independent Security?

A key question is whether P2’s resistance comes from the echo-canary signal, the semantic-alignment check, or their combination. We run an ablation with three configurations (90 trials each, same tasks/attacks/seeds):

The canary *does* provide independent security value: canary-only achieves ASR 0.03 (vs. 0.74 baseline), a 96% relative reduction, without the alignment check. However, all three configurations have Utility ≤ 0.04 . The combination does not recover Utility—it merely stacks two over-blocking signals. This refutes the original hypothesis that orthogonal multi-signal combination could achieve security *at acceptable utility cost*: in our end-to-end evaluation, P2’s Utility is 0.00, not the 0.43 reported by the earlier proxy-based measurement. The canary’s independent value is real but comes at full Utility cost.

VI. DESIGN PRINCIPLES AND FRAGILITY CONJECTURE

A. Single-Signal Fragility Hypothesis

Based on our empirical observations, we propose the following falsifiable hypothesis rather than a formal theorem, as the formal model of “signal” and “computability” requires assumptions about LLM internals that we cannot fully verify.

Conjecture 1 (Single-Signal Fragility, Empirical). *Let D be an IPI defense whose decision rule depends primarily on a single signal s that is observable to the attacker (i.e., the attacker can infer or approximate s from the prompt or from query feedback). Then an adaptive attacker with knowledge of the defense class can construct an injection that bypasses D while preserving the malicious instruction’s semantics, achieving ASR close to the no-defense baseline.*

a) *Scope and Assumptions.*: This hypothesis applies to defenses where (1) the signal is observable (e.g., delimiter text, encoding scheme), (2) the attacker knows the defense class and can craft a matched evasion strategy, and (3) the evasion suffix is additive and does not destroy the malicious instruction’s semantics. We do not claim a specific search complexity bound.

b) *Empirical Status: Not supported by our attacks.*: Our end-to-end experiments do *not* support this hypothesis with the attackers we implement. Input-encoding defenses achieve a capability-monotonic worst-case (best-of- k) Robust ASR of only 0.19–0.30 (static ASR 0.09–0.10), far below the 0.74 no-defense baseline. Moreover, the defense-matched signal-alignment suffix—the candidate meant to exploit the delimiter signal—achieves $\text{ASR} \leq 0.01$; the small worst-case increase over the static payload comes from a generic authority suffix, not from signal mimicry. We therefore cannot claim the delimiter signal is exploitable, and we retain the statement only

TABLE III

MAIN RESULTS ON AGENTDOJO WORKSPACE (GPT-4O-MINI, 3 SEEDS, 90 ATTEMPTS PER CELL). L0 = BENIGN UTILITY (NO ATTACK). L1 = STATIC IMPORTANT_INSTRUCTIONS ASR. MATCHED/GENERIC = ASR OF THE DEFENSE-MATCHED SUFFIX AND A TARGET-INDEPENDENT AUTHORITY SUFFIX. **ROBUST** = CAPABILITY-MONOTONIC WORST CASE max OVER THE L2 CANDIDATE SET (WHICH INCLUDES THE EXACT L1 PAYLOAD, MATCHED/GENERIC SUFFIXES, A LENGTH-MATCHED CONTROL, AND 3 PARAPHRASES); $\text{ROBUST} \geq \text{ASR}_{L1}$ BY CONSTRUCTION. ASR FROM SECURITY_FROM_TRACES, UTILITY FROM USER_TASK.UTILITY, ON REAL TOOL TRACES. PAIRED MCNEMAR p AND CLUSTER-BOOTSTRAP CIs FOR THE L1→ROBUST DIFFERENCE ARE IN APPENDIX A. EXCEPT SPOTLIGHTING (AGENTDOJO OFFICIAL), ROWS ARE OUR REIMPLEMENTATIONS.

Defense	Class	L0 Util↑	ASR _{L1}	ASR _{matched}	ASR _{generic}	Robust ASR↓
No Defense	—	0.90	0.63	0.52	0.71	0.86
Polymorphic	IE	0.90	0.10	0.00	0.27	0.29
Mixture-of-Enc.	IE	0.87	0.10	0.00	0.26	0.30
Spotlighting	IE	0.80	0.09	0.01	0.19	0.19
StruQ (prompt)	SQ	0.83	0.63	0.51	0.69	0.89
FATH	RC	0.03	0.00	0.00	0.00	0.00
Task Shield	RC	0.43	0.06	0.03	0.03	0.07
IPIGuard	RC	0.43	0.40	0.40	0.42	0.52
P1: Echo Canary	RC	0.03	0.02	0.00	0.00	0.04
P2: Orthogonal	RC	0.00	0.00	0.00	0.00	0.00
P3: Task Invariant	RC	0.50	0.42	0.37	0.39	0.50

IE = Input Encoding, SQ = Structured Query, RC = Runtime Checking.

TABLE IV

P2 ABLATION: CANARY VS. ALIGNMENT CONTRIBUTIONS (90 TRIALS EACH).

Configuration	ASR ↓	Utility ↑
Canary only	0.03 [0.01, 0.09]	0.00 [0.00, 0.04]
Alignment only	0.09 [0.05, 0.17]	0.04 [0.02, 0.11]
Combined (P2)	0.00 [0.00, 0.04]	0.00 [0.00, 0.04]

as a conjecture that our fixed candidate set fails to confirm; a search-based attacker might yet support or refute it.

B. Design Principles

We instantiate three design principles as minimal prototypes:

a) P1: Echo Canary.: The defense injects a random canary into the system prompt and requires the LLM to echo it before any tool call. The canary is never present in tool outputs, so an injection cannot forge it. *Result (end-to-end)*: ASR 0.03 [0.01, 0.09], Utility 0.00. The canary *does* reduce ASR substantially (from 0.74 baseline to 0.03), but Utility is 0.00 because the model frequently fails to echo the canary correctly, causing the defense to neutralize legitimate tool outputs. This prototype is intentionally minimal and *naive*: a more robust design would use a trusted monitor that holds the secret and verifies a MAC over the action, never exposing the secret to the LLM. Our P1 result demonstrates that the *naive* form (echo-the-canary) provides security but at full utility cost.

b) P2: Orthogonal Signals.: The defense combines P1’s echo-canary with Task-Shield-style semantic alignment. Evasion requires bypassing both checks. *Result (end-to-end)*: ASR 0.00 [0.00, 0.04], Utility 0.00. An ablation (Section V-G) disentangles the contributions: canary-only achieves ASR 0.03, alignment-only achieves ASR 0.09, and combined achieves

ASR 0.00. The canary provides independent security value (96% relative ASR reduction alone), but all configurations have Utility ≤ 0.04 . The combination does not recover Utility—it stacks two over-blocking signals. This refutes the original hypothesis that orthogonal multi-signal combination could achieve security at acceptable utility cost.

c) P3: Task Invariant (Relaxed).: The defense uses a tool-allow-list derived from the task’s expected tool set, blocking any action that calls a tool not in the allow-list. *Result (end-to-end)*: ASR 0.42 [0.32, 0.52], Utility 0.02. The allow-list blocks some injections (those that request off-list tools) but misses injections that use allowed tools maliciously (e.g., sending an email to an attacker address via the allowed email tool). The near-zero Utility reflects the defense’s restrictiveness (benign utility 0.50). This is itself a finding: static tool-allow-lists are insufficient for injection defense because they cannot distinguish malicious use of allowed tools from legitimate use.

d) P4: Least Privilege.: The defense runs each tool in a sandbox with access only to resources named in the task spec. We do not report a quantitative ASR for P4 because evaluating a containment defense by injection ASR is a *category error*: the sandbox’s goal is to limit blast radius, not to detect injections. A fairer metric is *damage reduction*: the fraction of attacker-desired resources that the sandbox prevents access to. Under this metric, P4 blocks access to non-allowed tools entirely, but cannot prevent malicious use of allowed tools (e.g., sending an email via the allowed email tool to an attacker address). P4 is excluded from the main table (Section V-F) because our implementation is a prompt-level simulation, not real OS-level sandboxing.

C. Implications

The principle prototypes reveal a nuanced picture. P1 (single canary signal) achieves ASR 0.03 but Utility 0.00—the canary

provides security but at full utility cost. P2 (orthogonal multi-signal) achieves ASR 0.00 but also Utility 0.00; the ablation confirms the canary contributes independent security value, but combining signals does not recover utility. P3 (tool-allow-list) achieves ASR 0.42 with Utility 0.02—moderate security, insufficient because it cannot detect malicious use of allowed tools. The fundamental tradeoff is that runtime-checking defenses that achieve low ASR tend to also destroy utility, either through aggressive blocking (P1, P2) or through incomplete coverage (P3). Input-encoding defenses, which do not block actions but reshape the context, achieve the best tradeoff by suppressing injected instructions while preserving the model’s ability to complete legitimate tasks.

VII. RELATED WORK

a) Indirect Prompt Injection.: Greshake et al. [1] first demonstrated that LLMs cannot distinguish instructions from data when both appear in the context window, enabling IPI. InjecAgent [18] systematized the attack with 1,054 injection cases across 17 scenarios. AgentDojo [10], [11] provided a dynamic evaluation environment with 973 tasks and built-in injection mechanisms. WebInject [19] extended IPI to multimodal web agents. AgentVigil [20] introduced black-box red-teaming for IPI. MUZZLE [21] proposed adaptive agentic red-teaming against web agents. Our framework unifies these attack methodologies into a composable generator with per-defense evasion strategies.

b) Defenses Against IPI.: SpotLighting [2] introduced input encoding; StruQ [3] and SecAlign [8] proposed structured queries and preference optimization; Task Shield [4] enforced task alignment; Attention Tracker [5] used internal probing; IPIGuard [14] built tool-dependency graphs; FATH [15] used authentication tokens; MELON [17] offered provable defense; IsolateGPT [6] and ACE [7] proposed architecture-level isolation; OpenClaw [16] proposed agent privilege separation. Our taxonomy organizes these into three evaluated classes (IE, SQ, RC) and evaluates them end-to-end under both static and adaptive attack.

c) Adaptive Attacks.: Zhan et al. [9] showed that adaptive attacks break SpotLighting, StruQ, and SecAlign. Our work extends this to 11 defenses evaluated end-to-end with native AgentDojo evaluators, comparing static (L1) vs. adaptive (L2) attacks under real tool execution. Chen et al. [22] showed that backdoor-augmented injection nullifies defense methods.

d) Concurrent Surveys.: Gulyamov et al. [23] provide a comprehensive review of prompt injection vulnerabilities and defense mechanisms. Geng et al. [24] survey attack methods, root causes, and defense strategies. Unlike these surveys, our work is an empirical SoK with a native adaptive attack framework and end-to-end evaluation using AgentDojo’s native security and utility evaluators; the surveys are taxonomic and do not conduct adaptive-attack evaluations.

e) LLM Agent Security.: Beyond IPI, recent work studies tool-selection attacks [25], cross-tool pollution, over-privileged tool selection, multi-agent injection propagation, and LLM agent system architectures. Our SoK focuses specifically on

the IPI attack-defense ecosystem and complements these broader surveys.

VIII. LIMITATIONS AND ETHICS

a) Limitations.: (1) We evaluate on a single backbone (GPT-4o-mini) and a single benchmark suite (AgentDojo workspace). The no-defense benign utility is 0.90, indicating the model can complete most tasks without attack, but future work should validate on stronger backbones and additional suites. (2) SecAlign [8] and the original fine-tuned StruQ [3] require model checkpoints incompatible with our API-based pipeline; our `struq_prompt` is a prompt-level approximation only. MELON [17], ACE [7], and IsolateGPT [6] require infrastructure not available in our framework. (3) We evaluate two attack types: static (`important_instructions`) and defense-matched adaptive (five evasion strategies). Other AgentDojo attacks may yield different patterns. (4) We use 3 seeds and 10 user tasks; wider coverage would tighten confidence intervals. (5) SpotLighting uses AgentDojo’s official delimiting implementation (high fidelity); other defenses are reimplemented from paper descriptions and may differ in prompt templates and threshold values.

b) Ethics Considerations.: This work studies attacks on LLM agents. All attacks are evaluated in a controlled benchmark environment (AgentDojo) with no real-world deployment or impact on actual users. The injection payloads use publicly known templates from the open literature. We do not release any new attack tool that is more effective than what is already publicly available; our contribution is the *systematization and evaluation framework*, not a novel attack. We disclose no real-world vulnerabilities. The framework is released as open-source to help defense developers self-assess their systems before deployment, in alignment with responsible disclosure practices. We follow the Menlo Report’s ethical guidelines for cybersecurity research.

c) Use of Generative AI.: Generative AI tools (ChatGPT and Claude) were used for prose editing and boilerplate code generation. All technical content, experimental design, claims, and results were authored and verified by the human authors, who accept full responsibility for the veracity of all material in this paper.

d) Literature Search Methodology.: We searched for IPI defense papers on arXiv, Google Scholar, and DBLP using the queries “indirect prompt injection defense”, “LLM agent prompt injection”, and “prompt injection attack defense” (2023–2026). Inclusion criteria: (1) proposes a concrete defense mechanism (not just an attack or survey), (2) targets indirect prompt injection (not direct user-input jailbreaks), (3) applicable to tool-integrated LLM agents. Exclusion criteria: (1) defenses for non-agent LLM applications (e.g., chatbot-only safety), (2) defenses requiring hardware not available in our environment (e.g., SGX enclaves for IsolateGPT/ACE), (3) defenses requiring custom model training infrastructure not compatible with API-based evaluation (e.g., SecAlign’s DPO pipeline). We identified 17 candidate defenses; 8 met all inclusion criteria and were evaluated, 4 were classified but not

evaluated (SecAlign, MELON, ACE, IsolateGPT), and 5 were excluded as out-of-scope (e.g., direct jailbreak defenses).

e) Implementation Fidelity.: Of the 7 literature baselines evaluated, Spotlighting uses AgentDojo’s official delimiting implementation (highest fidelity). StruQ is evaluated as a prompt-level approximation (`struq_prompt`) because the original requires a fine-tuned checkpoint. The remaining 5 (Task Shield, IPIGuard, FATH, Polymorphic Prompt, Mixture-of-Encodings) are reimplemented as native AgentDojo pipeline elements from the papers’ descriptions. Attention Tracker and P4 are excluded from quantitative evaluation due to fidelity limitations (see Section V-F). Our ASR and Utility numbers reflect our implementations, not the original authors’ code; we report this honestly and encourage future work to validate with official implementations.

f) Reproducibility.: All code, configurations, and results are available at the project repository (anonymized for double-blind review). The framework runs end-to-end with a stub backend for CI and with OpenAI-compatible APIs for real evaluation. Checkpointing supports resuming interrupted runs.

IX. CONCLUSION

We presented a systematization of knowledge on indirect prompt injection (IPI) defenses for LLM agents, evaluated end-to-end. Using AgentDojo’s native evaluators—measuring real tool-execution traces for security and real environment state for utility—we evaluated 10 defenses plus a no-defense baseline across three classes in 2,580 trials spanning benign (L0), static-attack (L1), and matched-suffix (L2) conditions. Our central contribution is that *end-to-end evaluation changes the ranking of IPI defenses*: input-encoding defenses achieve the best security-utility tradeoff, with a capability-monotonic worst-case (best-of- k) Robust ASR of 0.19–0.30 at benign utility 0.80–0.90, whereas a string-matching proxy reported ASR 1.00 for the same defenses on the same setting. Our Robust ASR is computed over a matched-suffix candidate set that always includes the static payload, so it is $\geq$ the static ASR (0.09–0.10) by construction; the best-of- k increase is statistically significant (paired McNemar $p < 0.005$) but comes from a generic authority suffix rather than the defense-matched suffix ($\text{ASR} \leq 0.01$). We explicitly do *not* claim to have built a signal-targeting adaptive attacker or that these defenses resist one—establishing genuine adaptive robustness would require optimization- or search-based attackers beyond our fixed candidate set. Second, by measuring benign utility separately (no-defense: 0.90), we show that 71% of the attack-condition utility loss is attack-induced failure, not a model capability limitation; defining benign over-blocking as $O_D = U_{\text{no-def}, L0} - U_{D, L0}$, input-encoding defenses over-block 0–0.10 while runtime-checking defenses over-block 0.47–0.90. Third, a P2 ablation shows the echo-canary provides independent security value (ASR 0.03 alone) but all signal combinations sacrifice utility entirely. These results, obtained on GPT-4o-mini and the AgentDojo workspace suite, demonstrate that end-to-end evaluation with native evaluators and benign baselines is essential for honestly characterizing IPI defense

tradeoffs; broader models, suites, and search-based attackers are important future work. We release our framework, native evaluation pipeline, and attack implementations as artifacts.

APPENDIX

Table V reports the L1 (static-payload) ASR and Utility with 95% Wilson confidence intervals, computed from the native end-to-end results. These are *marginal* proportions over 90 attempts per defense (10 user tasks $\times$ 3 injection tasks $\times$ 3 seeds).

a) Paired L1-vs-Robust analysis.: Because L1 and every L2 candidate reuse the same (user task, injection task, seed) units, the comparison is paired. We therefore do *not* rely on marginal Wilson intervals for the paired difference $\Delta = \text{Robust ASR} - \text{ASR}_{L1}$; instead we report an exact McNemar test on the discordant pairs and paired bootstrap confidence intervals for Δ , clustering the bootstrap by user task to account for the hierarchical structure (the same task recurs across seeds, so the 90 attempts are not independent). Because the L2 candidate set contains the exact L1 payload, $\Delta \geq 0$ by construction, and the McNemar test measures whether the set adds *additional* successes beyond L1. Table V reports the results (analysis code: `sok_ipl/native/monotonic_stats.py`). The best-of- k set significantly increases ASR over L1 for the input-encoding defenses, StruQ, IPIGuard, P3, and no-defense (McNemar $p < 0.02$, cluster-bootstrap CI excludes zero), and does not significantly increase it for FATH, Task Shield, P1, and P2 (which suppress ASR only by over-blocking; ?? 0a).

TABLE V
PAIRED L1-VS-ROBUST ANALYSIS (N=90 PER DEFENSE, 3 SEEDS).
ASR_{L1} = STATIC PAYLOAD; ROBUST = CAPABILITY-MONOTONIC WORST CASE OVER THE L2 CANDIDATE SET (INCLUDES L1). $\Delta = \text{ROBUST} - \text{ASR}_{L1}$ WITH 95% PAIRED-BOOTSTRAP CI; “CLUSTER CI” CLUSTERS BY USER TASK; p IS THE EXACT MCNEMAR TWO-SIDED p -VALUE.

Defense	ASR _{L1}	Robust	Δ [95% CI]	Cluster CI
Spotlighting	0.09	0.19	+0.10 [+0.04, +0.17]	[+0.02, +0.20]
Polymorphic	0.10	0.29	+0.19 [+0.11, +0.28]	[+0.08, +0.32]
Mixture-of-Enc.	0.10	0.30	+0.20 [+0.12, +0.29]	[+0.07, +0.36]
StruQ (prompt)	0.63	0.89	+0.26 [+0.17, +0.36]	[+0.14, +0.36]
FATH	0.00	0.00	+0.00 [+0.00, +0.00]	[+0.00, +0.00]
Task Shield	0.06	0.07	+0.01 [+0.00, +0.03]	[+0.00, +0.03]
IPIGuard	0.40	0.52	+0.12 [+0.06, +0.19]	[+0.06, +0.20]
P1: Echo Canary	0.02	0.04	+0.02 [+0.00, +0.06]	[+0.00, +0.07]
P2: Orthogonal	0.00	0.00	+0.00 [+0.00, +0.00]	[+0.00, +0.00]
P3: Task Invariant	0.42	0.50	+0.08 [+0.02, +0.13]	[+0.01, +0.16]
No Defense	0.63	0.86	+0.22 [+0.14, +0.31]	[+0.12, +0.33]

Table VI details the P2 ablation (90 trials per configuration, same tasks/attacks/seeds as the main experiment). The canary provides independent security value (ASR 0.03 alone vs. 0.74 baseline), but all configurations sacrifice Utility entirely.

To distinguish genuine defense over-blocking from attack-induced failure, we compare Task Shield’s benign (L0) and attack-condition (L1) utility with the no-defense baseline. Without attack, no-defense completes 90% of tasks (benign Utility 0.90); under attack, this drops to 19% (attack-condition

TABLE VI
P2 ABLATION: CANARY VS. ALIGNMENT CONTRIBUTIONS (N=90 EACH).

Configuration	ASR [95% CI]	Utility [95% CI]
Canary only	0.03 [0.01, 0.09]	0.00 [0.00, 0.04]
Alignment only	0.09 [0.05, 0.17]	0.04 [0.02, 0.11]
Combined (P2)	0.00 [0.00, 0.04]	0.00 [0.00, 0.04]

Utility 0.19). This 71-percentage-point drop is attack-induced failure, not model capability limitation. Task Shield’s benign over-blocking is $O_D = 0.90 - 0.43 = 0.47$: it blocks 47% of benign tasks even without any attack. Its attack-condition utility (0.02) is a further 0.41 below its benign utility (0.43); this additional drop reflects the joint effect of the attack and the defense. The key insight is that the L0 baseline is essential: without it, one might attribute Task Shield’s low attack-condition utility to the model, but the benign comparison shows the defense itself over-blocks nearly half of all benign tasks.

The prior version of this paper (proxy-based evaluation) included backbone comparisons (GPT-4o, Qwen2.5-7B), cross-benchmark validation (InjecAgent), adaptive-attack-strength sweeps, and SecAlign evaluation. These were conducted under the string-matching proxy methodology and produced results that *contradict* our native end-to-end findings (e.g., proxy reported SpotLighting R-ASR=1.00; native shows ASR=0.21). We have therefore *removed* these proxy-based results from this revision rather than report numbers we cannot support with end-to-end evidence. We plan to re-run backbone, cross-benchmark, and SecAlign evaluations using the native pipeline in future work.

Our evaluation uses AgentDojo v1.2.2’s native benchmark infrastructure. Each defense is implemented as a `BasePipelineElement` that integrates into AgentDojo’s `AgentPipeline` and `ToolsExecutionLoop`. The agent executes real tools (e.g., `send_email`, `get_calendar_events`), producing real function-call traces and environment state changes. Security is scored by `injection_task.security_from_traces(traces, pre_env, post_env)` and utility by `user_task.utility(model_output, pre_env, post_env)`. This eliminates the string-matching proxy used in prior work. All code is in `sok_ipl/native/pipeline_builder.py` and `scripts/run_bestofk.py`.

b) Capability-monotonic L2 candidate set.: For each unit $u = (\text{defense}, \text{seed}, \text{user task}, \text{injection task})$ we build a candidate set

$$\mathcal{A}_{L2}(u) = \{a_{L1}, a_{\text{matched}}, a_{\text{generic}}, a_{\text{control}}, a_{\text{para},0}, \dots, a_{\text{para},k-1}\},$$

where a_{L1} is the exact AgentDojo `important_instructions` payload (verified byte-for-byte), each other candidate is a_{L1} with an additive suffix inserted at a single fixed boundary after the goal (so the malicious goal is preserved verbatim), a_{generic}

is a target-independent authority suffix, and a_{control} is a semantically-inert filler suffix matched in character length to a_{matched} and derived deterministically by SHA-256 (isolating content effects from length). The per-unit outcome is $\text{success}(u) = \bigvee_{a \in \mathcal{A}_{L2}(u)} \text{security_from_traces}(a)$, and

$$\text{Robust ASR} = \text{mean}_u \max_{a \in \mathcal{A}_{L2}(u)} \text{ASR}(a) \geq \text{ASR}_{L1},$$

since $a_{L1} \in \mathcal{A}_{L2}(u)$. The FATH matched candidate embeds the real SHA-256(user task)[8] token; the P2 matched candidate attacks both the canary and the alignment check. The runner (`scripts/run_bestofk.py`) writes one attempt record per candidate and `sok_ipl/native/monotonic_stats.py` computes best-of- k , McNemar, and cluster bootstrap, asserting `l1_success  $\Rightarrow$  l2_set_success` for every unit.

The prior version of this paper included supplemental experiments (unknown-defense settings, no-feedback settings, cross-defense transfer, threshold sweeps, and held-out defense validation) conducted under the string-matching proxy methodology. Because these proxy results *contradict* our native end-to-end findings (e.g., proxy reported SpotLighting ASR=1.00; native shows ASR=0.21), we have removed them from this revision rather than present unsupported numbers. We summarize what was removed and what remains:

c) Removed (proxy-based, not re-run natively).: (1) Settings A/B/C (unknown-defense, no-feedback, cross-defense transfer)—all used proxy ASR. (2) Threshold sweep for Task Shield and Attention Tracker—used proxy FPR/USR metrics; the native pipeline does not use the same threshold mechanism. (3) Held-out defense validation via SecAlign—used proxy ASR on Mistral-7B. (4) Query/cost analysis referencing the 3,900-trial proxy evaluation.

d) Retained (native end-to-end).: (1) Main results table (Table III): 10 defenses plus no-defense across L0/L1/L2 (330 benign + 990 static + 990 matched-suffix = 2,310 main trials), native `security_from_traces` + `user_task.utility`. (2) P2 ablation (Section V-G, ?? 0a): 270 trials, three configurations, giving 2,580 total. (3) Over-blocking diagnosis (?? 0a): benign L0 baseline comparison. (4) Statistical analysis (Appendix A): marginal Wilson CIs plus paired McNemar and cluster bootstrap for L1-vs-L2 differences.

e) Capability-monotonic attack set.: Our L2 condition is a candidate set that always contains the exact static (L1) payload plus a defense-matched suffix, a generic authority suffix, a length-matched control, and k paraphrases; we report the worst-case (best-of- k) Robust ASR = $\max(\text{ASR}_{L1}, \text{ASR}_{L2\text{-set}})$, which is $\geq \text{ASR}_{L1}$ by construction (?? 0a). The length-matched control isolates whether a matched suffix’s effect is due to content or merely to added length.

f) Planned.: A matched-vs-mismatched transfer matrix and search-/optimization-based adaptive attackers (beyond our fixed candidate set) are planned future work; the native pipeline infrastructure (?? 0a) supports them, and the current

constraint is API cost and time. Broader coverage (additional backbones and AgentDojo suites) is likewise future work.

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